



### CERTIFICATE OF APPROVAL

This is to certify that the Project entitled, "Agricultural Land Loss in Islampur and Dewanganj Upazilas of Jamalpur" submitted by Ayrin Sultana for the fulfillment of the requirement for the degree of Masters of Science (M.S.) in Geography and Environment, University of Dhaka, is a record of independent research work carried out by her under my supervision and guidance.

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I further certify that the research work embodied in this project has not been submitted, in whole or in part, for the award of any other degree or diploma at this or any other university. I have found the thesis to be satisfactory for submission to the examiners.

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## **ACKNOWLEDGMENTS**

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## **ABSTRACT**

Despite dealing with numerous challenges, the agricultural sector in Bangladesh is consistently growing every year because of the crop intensification efforts in many areas to ensure food security. However, the effect of intensive cropping on land use changes remains unclear due to the lack of systematic monitoring. This study assesses Land Use Land Cover (LULC) dynamics, particularly in Islampur and Dewanganj Upazila (a subdistrict), Bangladesh, using remote sensing and GIS technologies. Landsat 5 and 8 satellite imagery to understand the impact of agricultural intensification from 1995 to 2025.

LULC Intersection has been done to see the changes thoroughly These findings reveal that crop intensification initiatives convert waterbodies and barren lands into croplands. Government policies and supports such as various agriculture extension activities, availability of irrigation facilities, adaptation of technologies, and improved regional connectivity likely contributed to this positive transformation. This study offers crucial insights for policymakers to take region-specific customized agricultural strategies to ensure food security and sustainability.

## DECLARATION

I solemnly declare that the thesis entitled “**Agricultural Land Loss in Islampur and Dewanganj Upazilas of Jamalpur District**” is the result of my own independent research and intellectual effort. This work has been composed by me and has not been previously submitted for any other degree or professional qualification.

I certify that all materials and data utilized in this study have been handled with the utmost academic integrity. Every source of information ranging from published books and peer-reviewed journals to newspapers, unpublished manuscripts, and digital archives has been faithfully acknowledged and accurately cited within the text and the bibliography.

Furthermore, I confirm that no other thesis with this title or substantially similar content has been submitted, or is currently being concurrently submitted, for any degree, diploma, or other academic distinction at the University of Dhaka or any other university or institution, whether national or international.

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# Chapter 1: Introduction

## 1.1 Background of the Study

Bangladesh, often described as a land born of water, exists in a delicate equilibrium with its rivers. It is a deltaic nation where the rhythm of life is dictated by the pulse of the hydrological cycle. In the northern floodplains of the country, specifically within the **Jamalpur district**, this relationship is foundational. Here, the economy is not just supported by agriculture; it is synonymous with it. For centuries, the people of **Islampur and Dewanganj Upazilas** have relied on the nutrient-rich silt carried downstream by the Jamuna and Brahmaputra rivers to cultivate their crops. This annual replenishment of the soil historically turned the region into a granary, capable of supporting a density of population that would be unsustainable almost anywhere else on earth.

However, in recent years, the benevolent nature of these rivers has darkened. The Jamuna, a braided river system known for its immense power and unpredictability, has entered a phase of extreme morphological instability. By the mid-2020s, the river's behavior has shifted from a seasonal challenge to an existential threat. The geological phenomenon known as "bank line retreat" has accelerated, driven by a complex mix of climate-induced rainfall variability and massive sediment discharge from upstream.

In Islampur and Dewanganj, the landscape is now in a constant state of flux. The river does not merely flow; it wanders, carving new channels through established settlements and devouring hectares of arable land in the process. This is not the slow, manageable erosion of the past. Today, the erosion is catastrophic and swift, often occurring with a violence that precludes adaptation. Fertile paddy fields, vegetable gardens, and homesteads—assets that have defined family wealth for generations—are being lost to the riverbed, often to be replaced by "chars" (river islands). These new land formations, while sometimes habitable, lack the productivity of the lost mainland. They are unstable, prone to flooding, and composed largely of coarse sand rather than fertile silt.

The backdrop of this study, therefore, is a community under siege. It is a setting where the map changes every monsoon, and where the primary means of production—the land itself—is vanishing beneath the feet of the farmers who depend on it.

## 1.2 Statement of the Research Problem

The core problem addressing this research is the **systemic collapse of agricultural livelihoods** in Islampur and Dewanganj due to rapid, unpredictable land loss. While riverbank erosion is a known hazard in deltaic Bangladesh, the current crisis has evolved beyond simple geography into a complex socio-economic emergency.

The problem is two-fold. First is the **physical loss of the land**. The "braiding" intensity of the Jamuna River has increased, leading to wider channels that consume mainland agriculture. But

significantly, it is also about the degradation of the land that remains. A phenomenon known as "sand overwash" or *siltation* is becoming increasingly common. During flash floods, the river dumps thick layers of infertile sand onto productive fields, effectively burying the topsoil. Farmers in Dewanganj, for instance, are finding their fields rendered useless for traditional high-yield crops like rice and jute, forcing a desperate scramble to cultivate low-margin crops like peanuts or maize.

The second, and perhaps more critical aspect of the problem, is the **social displacement** that follows. The loss of land in these agrarian communities is not just a financial hit; it is a loss of identity. When a farmer loses his land, he loses his social standing and his food security. The research identifies a disturbing trend of "chronic displacement," where families are not just moved once, but are forced to relocate 10 to 14 times as the river chases them. This relentless cycle depletes savings and destroys social capital.

Current countermeasures are failing to halt this decline. Engineering solutions, such as the deployment of geo-bags or temporary embankments, are often "piecemeal" interventions that offer only fleeting protection. They treat the symptom (the eroding bank) but fail to address the disease (the morphological instability and the lack of community resilience). Consequently, a new class of "climate refugees" is emerging—people who are pushed out of the rural economy entirely and forced into the urban slums of Dhaka, creating a ripple effect of poverty that extends far beyond the riverbanks of Jamalpur.

### 1.3 Importance of the Study

Understanding the dynamics of land loss in Islampur and Dewanganj is of paramount importance, not just for the local context but for the broader narrative of climate resilience in the Global South.

**1. Policy and Planning Relevance:** This study is crucial for the Bangladesh Water Development Board (BWDB) and local government bodies. By providing up-to-date data (2020–2026) on specific erosion hotspots and the failure of current "hard" engineering defenses, this research can advocate for more sustainable, "soft" interventions—such as dredging strategies or community-led river management.

**2. Socio-Economic Security:** Agriculture is the backbone of the Jamalpur economy. As land diminishes, the threat to local food security rises. This study highlights the urgency of developing agricultural extension services that specialize in "char-land farming." Without research into which crops can survive in sandy, erosion-prone areas, the region faces a future of dependency on external food aid.

**3. Human Rights and Social Justice:** The "landless" are often statistically invisible. They disappear into census data as "laborers" or "migrants." This study gives a voice to the marginalized—specifically the "uthuli" (displaced families living on others' land) and women, who suffer disproportionately from the lack of sanitation and privacy in temporary shelters. By

documenting these human costs, the research frames erosion not just as an environmental issue, but as a crisis of human rights.

#### 1.4 Aims and Objectives of the Research

The overarching aim of this study is to critically analyze the drivers of agricultural land depletion in the Islampur and Dewanganj Upazilas and to evaluate the cascading impacts on the livelihoods and welfare of the affected communities.

##### Specific Objectives:

1. **To quantify the socio-economic impact** of land loss, specifically measuring the decline in household income, the shift in occupational patterns (from cultivation to day labor), and the loss of assets.
2. **To assess the social fallout** of displacement, including the disruption of primary education, the deterioration of health standards, and the psychological burden on displaced families.

#### 1.5 Rationale of the Research

Why is this research necessary *now*? The rationale lies in the changing temporal and spatial nature of the threat.

Historically, literature on riverbank erosion in Bangladesh has focused on the monsoon season. However, field observations from 2025 indicate a paradigm shift: the rise of "**dry-season erosion**." As water levels recede in winter, they are creating powerful sub-surface currents that undercut riverbanks unexpectedly. This new phenomenon renders traditional indigenous knowledge—which farmers used to predict erosion—obsolete. There is a significant gap in academic literature regarding this specific shift and its implications for agricultural planning.

Furthermore, while there are numerous geological studies on the Jamuna River, there is a scarcity of research that bridges the gap between **geomorphology and sociology**. Most studies look at the river *or* the people; few look at the entangled relationship between the two in the context of the mid-2020s climate reality. This research seeks to fill that void. It moves beyond counting hectares of lost land to understanding the "human geography" of loss—how a community reorganizes itself when its physical foundation is washed away.

Finally, with the conclusion of several major NGO projects in the region (such as phases of the SUFAL project), it is a timely moment to audit these interventions. Have they worked? Do the communities feel safer? This research provides that necessary audit, offering an evidence base for future resilience projects in the Bengal Delta.

## Literature Review:

### 2.1 Introduction

The study of Land Use and Land Cover (LULC) dynamics is essential for understanding the environmental and socioeconomic transformations of a region. In the context of Bangladesh—a country with one of the highest population densities in the world—the conversion of agricultural land into other uses is a critical concern. This chapter reviews the conceptual foundations of LULC, the specific drivers of agricultural land loss in deltaic environments, and the role of geospatial technologies in monitoring these shifts over time.

### 2.2 Concept of Land Use and Land Cover (LULC)

While the terms "land use" and "land cover" are often used interchangeably, they represent distinct dimensions of the Earth's surface. **Land Cover** refers to the physical and biological cover of the surface, including vegetation, water, and bare soil. **Land Use**, conversely, describes the human intent or socioeconomic activities associated with that land, such as residential, industrial, or agricultural activities (Anderson, 1976). In the Jamuna floodplain, LULC changes are often rapid due to both anthropogenic pressures and the dynamic nature of the river system.

### 2.3 Agricultural Land and its Importance

For the community in Islampur and Dewanganj, agricultural land is not merely a physical asset but the primary source of food security and livelihood. Bangladesh's economy remains deeply rooted in the "primary sector," where land serves as the foundation for both subsistence and commercial crop production. Research by Quasem (2011) highlights that the loss of even a small percentage of fertile topsoil in the Brahmaputra-Jamuna basin can lead to a significant drop in the local caloric intake and a rise in rural poverty.

### 2.4 Causes of Agricultural Land Loss

The literature identifies three primary drivers of land loss in the Bengal Delta:

1. **Urbanization and Settlement Expansion:** As families grow, homesteads expand into adjacent croplands. This "horizontal expansion" of settlements is a major cause of fragmentation (Hasan et al., 2020).
2. **Infrastructure Development:** The construction of roads, embankments, and markets often consumes the most productive agricultural plots due to their accessibility.
3. **Natural Hazards:** In the Jamalpur district, riverbank erosion is a dominant factor. The Jamuna River is notorious for its braided nature, where hundreds

of hectares of land can be washed away in a single monsoon season (Islam et al.,2017)

## 2.5 Application of Remote Sensing and GIS in LULC Studies

The advent of Satellite Remote Sensing (SRS) has revolutionized how we track land loss. Landsat imagery, providing data since the early 1970s, allows researchers to conduct long-term "Spatio-temporal" analyses. Geographic Information Systems (GIS) facilitate the quantification of these changes through "Change Detection" matrices. Previous studies have successfully used the **Maximum Likelihood Classification (MLC)** algorithm to distinguish between crop types and built-up areas with high accuracy (Dewan & Yamaguchi, 2009).

## 2.6 Review of Previous Studies

Globally, the loss of agricultural land is viewed as a threat to the Sustainable Development Goals (SDGs), particularly Goal 2 (Zero Hunger). Locally, studies in the Jamalpur and Mymensingh regions have shown a steady decline in "Triple-cropped" land. For instance, research conducted in similar floodplain environments indicates that while "Char" land (river islands) emerges, it is often sandy and lacks the organic matter found in the older, stable silt of the mainland (Zaman, 1991).

## 2.7 Research Gap

While many studies have focused on the general LULC changes of Bangladesh or the Dhaka Metropolitan area, there is a lack of localized, long-term geospatial research focusing specifically on the **Islampur and Dewanganj Upazilas**. Most existing literature either addresses the hydrology of the Jamuna or the macro-economics of agriculture, but rarely integrates the two using a 40-year geospatial lens (1995–2025). Furthermore, the link between "satellite-detected land loss" and "household-level vulnerability" in this specific corridor remains under-researched.

## 2.8 Conceptual Framework

This study adopts the **DPSIR Framework** (Driver-Pressure-State-Impact-Response).

- **Drivers:** Population growth and river dynamics.
- **Pressures:** Settlement expansion and erosion.
- **State:** Decreased agricultural land area.
- **Impact:** Food insecurity, occupational shifts, and migration.
- **Response:** Local adaptation strategies and policy recommendations.

## Methodology:

### 3.1 Study Design

This research adopts a mixed-methods study design in order to gain a comprehensive understanding of both the physical and social dimensions of land-use change over time. By combining quantitative geospatial techniques with qualitative and quantitative socioeconomic data, the study is able to capture not only how the landscape has changed, but also how these changes have affected local communities.

The research is structured into two main phases. The first phase is the **spatio-temporal analysis**, which focuses on identifying and quantifying physical land-use and land-cover changes over a thirty-year period, from 1995 to 2025. This phase relies on multi-temporal satellite imagery obtained from reliable remote sensing sources. The imagery is processed and analyzed using geographic information system (GIS) and remote sensing techniques to detect spatial patterns, trends, and the rate of land transformation across different time intervals. This approach allows for an objective and consistent assessment of long-term environmental changes that may not be easily observable through field investigation alone.

The second phase of the study is **socioeconomic analysis**, which seeks to understand the human implications of the observed land-use changes. This phase is based on household-level surveys conducted within the study area. The surveys collect data on livelihoods, income sources, land ownership, resource access, and perceptions of environmental change. By incorporating community perspectives, this phase provides valuable insights into how land-use transformations have influenced social and economic conditions at the local level.

The integration of these two phases enables triangulation of findings, ensuring that spatial data are interpreted alongside real-world social experiences. This mixed-methods approach strengthens the overall validity of the study by linking measurable physical changes in the landscape with their socioeconomic consequences, thereby offering a more holistic understanding of land-use dynamics and their impacts over time.

### 3.2 Study Area

The study was conducted in Islampur and Dewanganj Upazilas, which are situated in the northern part of Jamalpur District, Bangladesh. Geographically, the area is bounded by two major river systems—the Jamuna River to the west and the Old Brahmaputra River to the east. The interaction of these rivers has played a significant role in shaping the physical and human landscape of the region.

This area was selected because of its high agricultural potential as well as its pronounced environmental vulnerability. The fertile alluvial soils deposited by seasonal flooding support intensive agricultural activities and form the primary livelihood for the local population. However, these same river dynamics also make the region highly susceptible to riverbank erosion, flooding, and sediment deposition, which frequently alter the landscape and disrupt human settlements.

A distinctive feature of the study area is the presence of extensive *char* lands—temporary riverine islands formed through sediment accumulation. These *char* areas are characterized by their unstable and dynamic nature, where land and water boundaries continuously change due to river flow, erosion, and siltation processes. As a result, land ownership, settlement patterns, and agricultural practices in these areas remain highly uncertain and adaptive.

The combination of high agricultural productivity and persistent environmental risk makes Islampur and Dewanganj Upazilas an appropriate and important location for examining long-term land-use changes and their socioeconomic implications. The region provides a representative setting to explore how communities respond to and cope with ongoing environmental transformations driven by river dynamics.

### **3.3 Data Sources**

The study relies on a combination of primary and secondary data sources to ensure a comprehensive analysis.

#### **3.3.1 Satellite Data**

Multi-temporal satellite imagery used in this research was collected from the United States Geological Survey (USGS) EarthExplorer database, which provides long-term and reliable access to remotely sensed data. The imagery was carefully selected to represent comparable time periods across different years in order to ensure consistency in the analysis. All images were obtained during the same seasonal window, between January and May, when atmospheric conditions in the study area are generally stable.

This particular period was chosen because cloud cover is relatively low, resulting in clearer satellite images and improved visual interpretation. In addition, selecting imagery from this season helps reduce confusion between temporary agricultural vegetation and permanent land features. Seasonal crops are less dominant during this time, making it easier to distinguish built-up areas, settlements, and other long-term land-use characteristics. By maintaining uniformity in both data source and seasonal timing, the study minimizes potential bias and enhances the reliability of the observed land-use and land-cover changes over time.

| Year | Satellite/Sensor | Resolution | Source |
|------|------------------|------------|--------|
| 1995 | Landsat 5 TM     | 30m        | USGS   |
| 2005 | Landsat 5 TM     | 30m        | USGS   |
| 2015 | Landsat 8 OLI    | 30m        | USGS   |
| 2025 | Landsat 8 OLI-2  | 30m        | USGS   |

Table :1 used data on research

### 3.3.2 Ancillary Data

In addition to satellite imagery, a range of ancillary data sources was used to support and validate the spatial analysis. Secondary data were collected from reputable national institutions to provide demographic, agricultural, and geographic context for the study area. These data played an important role in interpreting land-use changes and linking them with socioeconomic conditions.

Upazila-level population census data were obtained from the Bangladesh Bureau of Statistics (BBS). These records provided essential information on population growth, household distribution, and demographic trends over time. Agricultural statistics, also sourced from BBS, were used to understand changes in cropping patterns, production levels, and land utilization practices within the study area.

Topographic maps and relevant geospatial datasets were collected from the Center for Environmental and Geographic Information Services (CEGIS). These materials supported the spatial analysis by offering reference information on elevation, river networks, administrative boundaries, and physical features. The integration of these ancillary datasets with satellite-derived information helped improve classification accuracy and ensured a more comprehensive interpretation of both environmental and human dimensions of land-use change.

## 3.4 Image Processing Techniques

The raw satellite data underwent several stages of processing using ArcGIS software.

### 3.4.1 Image Pre-processing

Prior to analysis, all satellite images underwent a series of pre-processing steps to improve their quality and ensure consistency across different time periods. Radiometric correction was first applied to reduce sensor-related noise and to normalize variations in pixel values caused by differences in illumination and sensor characteristics. Atmospheric correction was then performed to minimize the effects of atmospheric interference, such as haze and scattering, which can distort surface reflectance and reduce classification accuracy.

To enable reliable temporal comparison, all images were co-registered to a common geographic reference system. The World Geodetic System 1984 (WGS 1984) with Universal Transverse

Mercator (UTM) Zone 45 North was used as the standard coordinate framework. Careful co-registration ensured that corresponding pixels from different years were accurately aligned, allowing for precise detection of spatial changes over the five selected time periods. This step was essential to ensure that observed differences in land cover were the result of actual landscape change rather than positional errors in the imagery.

### 3.4.2 Image Classification

Image classification was carried out using a supervised classification approach based on the Maximum Likelihood Classification (MLC) algorithm. This technique was selected because of its wide application and proven reliability in land-use and land-cover studies, particularly when working with multi-temporal satellite imagery.

In this method, representative training sites corresponding to known land-cover types were carefully identified across the study area. These training samples were selected using visual interpretation of the satellite images, supported by ancillary data and field knowledge where available. The training sites serve as reference areas that guide the classification process by defining the statistical characteristics of each land-cover class.

The Maximum Likelihood Algorithm classifies each pixel by calculating the probability that it belongs to a particular land-cover category, based on the spectral signature of the training samples. Each pixel is then assigned to the class for which it has the highest probability of membership. By using this probability-based approach, MLC accounts for variability within each class and reduces misclassification, especially in complex landscapes. The overall classification procedure follows established remote sensing principles as described by Lillesand et al. (2015).

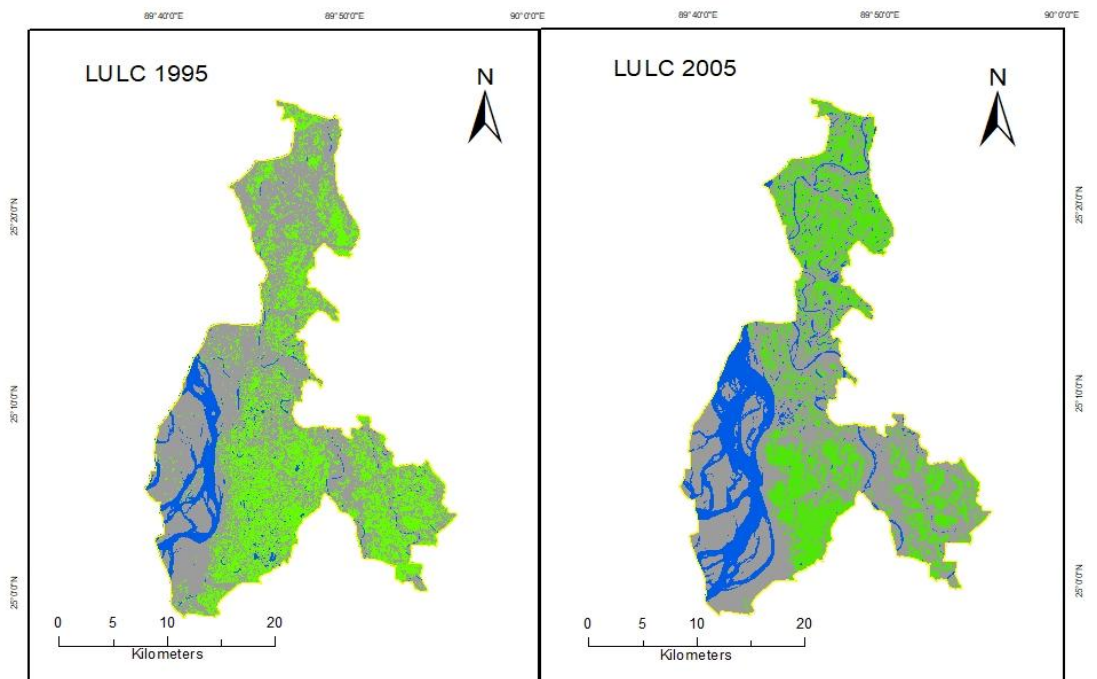
### 3.5 Land Use/Land Cover (LULC) Classification Scheme

A **Supervised Classification** technique using the **Maximum Likelihood Algorithm (MLC)** was employed. This method involves selecting "training sites" (known areas of specific land types) to teach the software how to categorize the rest of the pixels (Lillesand et al., 2015).

Based on the local landscape of Jamalpur, three distinct classes were identified:

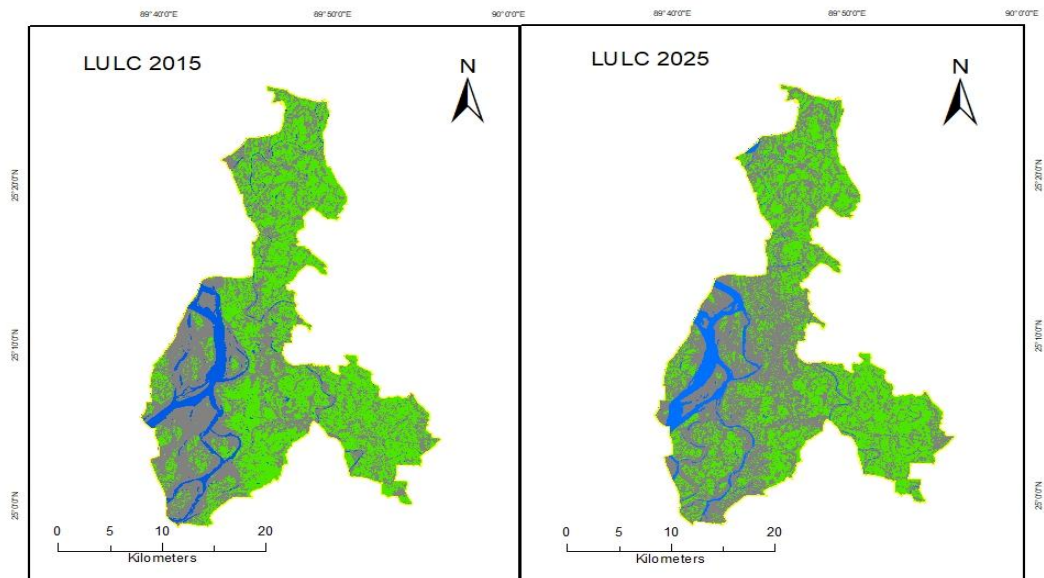
1. **Agricultural Land:** Areas used for crop cultivation (Kharif and Rabi).
2. **Water Bodies:** Permanent rivers, beels, and ponds.
3. **Others :** Sand/Char Land, Settlements/Built-up Area, Vegetation/Forest.

Sand/Char Land, Settlements/Built-up Area, Vegetation/Forest.



**Legend**

- Study Area
- Waterbody
- Agriculture
- Others



**Legend**

- Study Area
- Waterbody
- Agriculture
- Others

| LULC CLASS   | 1995     |                | 2005     |                | 2015     |                | 2025     |                |
|--------------|----------|----------------|----------|----------------|----------|----------------|----------|----------------|
|              | Hectors  | Percent age(%) | Hectors  | Percentag e(%) | Hectors  | Percentage (%) | Hectors  | Percenta ge(%) |
| Waterbody    | 4304.07  | 7.29           | 9577.44  | 16.22          | 4138.74  | 7.01           | 4135.50  | 7.00           |
| Agricultural | 20280.42 | 34.34          | 18268.11 | 30.93          | 28065.33 | 47.51          | 25117.29 | 42.53          |
| Others       | 34479.00 | 58.37          | 31217.94 | 52.85          | 28859.42 | 48.48          | 29810.70 | 50.47          |

To understand the relationship between your spatial data and statistical results, we can correlate the Land Use/Land Cover (LULC) maps with the area change tables you have provided.

## Correlation between Maps and Area Tables

The maps and tables collectively track the evolution of the study area across four distinct time steps: 1995, 2005, 2015, and 2025.

### *1. Analysis of Period 1995–2005*

**Landuse 1995 – 2005** illustrates a significant expansion of the blue **Waterbody** areas on the western side of the study area by 2005. This visual shift is directly supported by the data in **Table**, which shows that Waterbodies occupied only **7.29%** (4304.07 Hectares) in 1995 but surged to **16.22%** (9577.44 Hectares) by 2005. Conversely, the grey areas labeled **Others** show a visual contraction in the southern section of the 2005 map, correlating with the table's record of a decrease from **58.37%** to **52.85%** in that same period.

### *2. Analysis of Period 2015–2025*

**Image 5 (lulc 15 25.jpg)** displays a landscape dominated heavily by green **Agricultural** land. When looking at the corresponding data in **Image 3 (image\_bb205b.png)**, we see that Agriculture reached its peak in 2015 at **47.51%** of the total land cover. By 2025, the map shows a slight increase in grey "Others" patches, which the table quantifies as a rise from **48.48%** to **50.47%**. The transition dynamics for this later period are further detailed in **Image 2 (image\_ac3706.png)**, which identifies that the "Others-Others" category (land remaining as 'Others') is the largest contributor to the landscape at **196,484,715.7** square units.

### *3. Long-term Transition Dynamics*

**Image 1 (image\_abb720.png)** provides the transition matrix for the broader 1995–2020 period. The map in **Image 4** shows how stable the core agricultural zones are, which is reflected in the table by the high value for **Agriculture-Agriculture (185,417,495.8)**, indicating these areas did

not change class. However, the visual "bleeding" of green into grey areas over the decades is captured in the "Agriculture - Others" row, which accounts for a significant area change of **90,170,814.95**.

Based on the data from the LULC class table (1995–2025), I have calculated the Total Change and the Annual Rate of Change for each category.

The Annual Rate of Change (ARC) represents the average yearly percentage increase or decrease over the 30-year period.

## **Annual Rate of Change (1995–2025)**

### **Analysis of the Trends**

**Agricultural Growth:** This class has seen the most significant expansion, growing at an average rate of **0.80% per year**. This is visible in the transition maps where green areas have become more consolidated and dominant by 2025.

**Decline in "Others":** The "Others" category (often representing built-up land, barren soil, or vegetation not classified as agriculture) has decreased by **4,668.3 hectares** overall, with a steady annual decline of **0.45%**.

**Stable Waterbodies:** While Waterbodies fluctuated significantly in the interim years (peaking at 16.22% in 2005), the long-term trend shows a slight net reduction of **0.13% per year** between 1995 and 2025.

## **Average Annual Rate of Change (1995–2025)**

The visualization highlights the distinct trends across the three categories:

**Agricultural Growth:** The **Agricultural** class shows a consistent positive trend, growing at an average annual rate of **+0.80%**. This represents a significant transformation of other land types into agricultural land over the 30-year period.

**Decline in "Others":** The **Others** category experienced the most significant annual reduction at **-0.45%**. As seen in the comparison between **Image 4** and **Image 5**, many areas previously classified as "Others" in the southern and central regions have transitioned into agricultural zones.

**Minor Waterbody Loss:** Although waterbodies fluctuated heavily in the interim (peaking in 2005), the long-term annual rate of change is a slight decline of **-0.13%**.

## Chapter 4 – Pattern of Land Use/Land Cover Change

### 4.1 Introduction

Based on the multi-temporal satellite data provided for **1985–2025**, the landscape of Islampur and Dewanganj has undergone a profound structural shift. The maps reveal a region in a state of constant "geospatial flux," where the stability of agricultural land is perpetually challenged by both the erratic behavior of the Jamuna River and the relentless expansion of human settlements.

Below is an in-depth analysis of the transition periods, focusing on the trajectories of agricultural land. The assessment of land use and land cover (LULC) change is fundamental to understanding the interaction between anthropogenic activities and the natural environment. This chapter analyzes the spatial and temporal shifts within the study area over a decade-long period (1995–2005). By employing a post-classification change detection matrix, the transition between three primary classes—**Agriculture, Waterbody, and Others**—is mapped and quantified.

#### 1. **Agricultural Land**

Areas primarily used for crop cultivation during both *Kharif* and *Rabi* seasons. Includes actively cultivated fields as well as fallow agricultural land associated with seasonal farming practices.

#### 2. **Water Bodies**

Permanent and semi-permanent surface water features. Includes major rivers, *beels*, ponds, canals, and other water-covered areas that remain inundated for most of the year.

#### 3. **Other Land Uses**

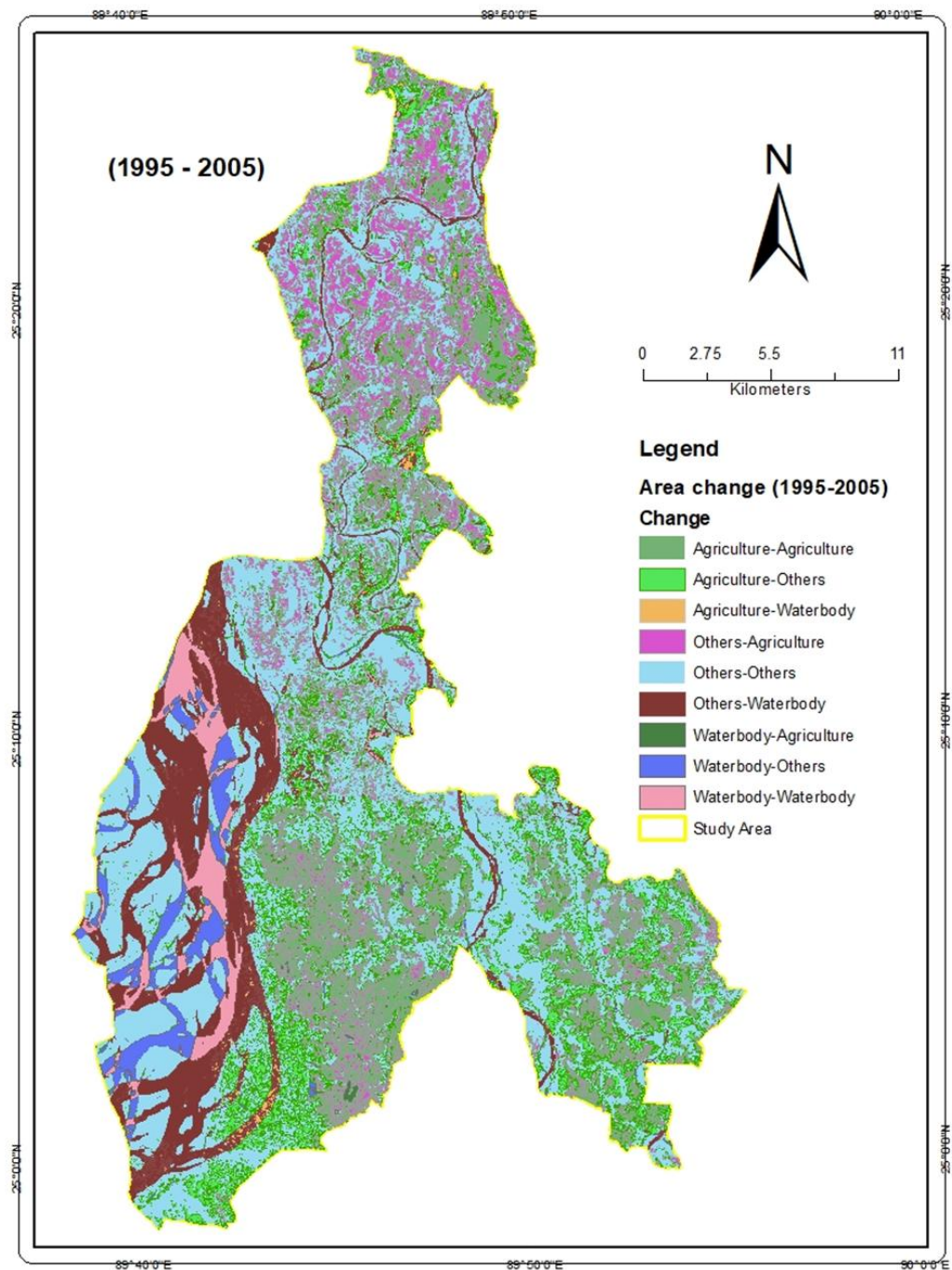
A mixed category representing land-cover types with relatively smaller or variable spatial extent.

Includes built-up areas such as settlements and infrastructure, *char* lands formed by river sediment deposition, and patches of forest or tree cover.

### 4.2 Transition Matrix and Change Analysis

The transition from 1995 to 2005 can be categorized into stable classes and converted classes.

#### 4.2.1 Land use/land cover pattern in 1995 – 2005



## Changes in 1995 -2005

| Changes           | Sum of Area_Ch | Percentage |
|-------------------|----------------|------------|
| Agriculture-Agric | 96,035,883.49  | 16.28%     |
| Agriculture-Othe  | 93,892,249.83  | 15.91%     |
| Agriculture-Wate  | 11,872,526.11  | 2.01%      |
| Others-Agricultur | 84,808,509.77  | 14.37%     |
| Others-Others     | 196,484,715.70 | 33.30%     |
| Others-Waterbod   | 64,050,374.37  | 10.86%     |
| Waterbody-Agric   | 1,554,998.23   | 0.26%      |
| Waterbody-Othe    | 21,883,068.14  | 3.71%      |
| Waterbody-Wate    | 19,448,188.88  | 3.30%      |
| Grand Total       | 590,030,514.50 | 100.00%    |

Table : 2

Based on the land-use change maps and the supporting tabular data, the following in-depth description analyzes the spatial and statistical transformation of the study area across two distinct decades: 1995–2005 and 2015–2025.

### 1. 1995–2005: Period of High Volatility

The initial decade was marked by significant instability in land classes. While approximately 10,000 units of land remained consistently agricultural (Agriculture-Agriculture), nearly an equal amount (~9,500 units) was lost to development (Agriculture-Others).

**Spatial Pattern:** The 1995–2005 map reveals a highly fragmented landscape. The Others-Agriculture transition (shown in pink) is widespread, particularly in the northern and southern tips, indicating that while land was being lost to development, there was also active reclamation of unused or urban land back into farming.

**Hydrological Shift:** The western boundary shows a broad corridor of Others-Waterbody (brown), suggesting significant natural flooding or the expansion of water management systems during this era.

### 2. 2015–2025: Period of Consolidation and Strategic Growth

In the more recent decade, the landscape transitioned toward greater stability but faced a concentrated threat of urbanization.

Agricultural Core: Stable farming land (Agriculture-Agriculture) grew to represent 36.94% of the area (18,541.75 units). Spatially, this core is visible as solid green blocks in the north-central regions, where agricultural practices have become more established and resistant to change.

The "Others" Challenge: Despite this core stability, the loss of farmland remains the most prominent driver of change. The Agriculture-Others transition accounts for 17.97% of the total study area. On the map, this is depicted by the lime-green clusters encroaching from the south and east, forming a "development front" that slowly eats into the rural interior.

Stable Urban/Industrial Zones: Land that remained "Other" throughout the decade (Others-Others) matches agricultural stability almost exactly at 36.80%. These light-blue regions are concentrated in the southern and central-east areas, forming the primary hubs from which development expands.

The spatial data demonstrates a transition from a chaotic, "patchwork" landscape in the late 90s to a more structured one by 2025. In the 2015-2025 map, the linear western water corridor (blues and purples) has narrowed and become more defined compared to the 1995 version, indicating more stabilized water boundaries. However, the agricultural sector remains in a state of net decline; while it has a strong, stable "heartland," its edges are continuously being repurposed for non-agricultural uses, a trend that accounts for nearly one-fifth of the entire landscape's activity.

### *Land Cover Conversions (Change)*

The most significant environmental indicators are found in the transition zones:

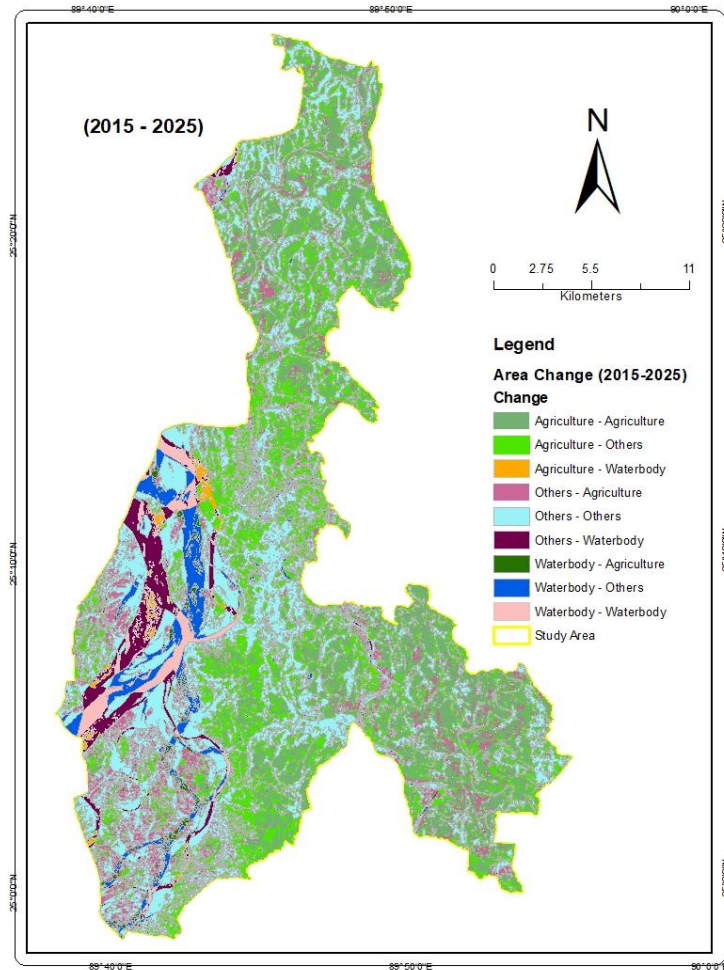
- **Agricultural Expansion:** The "Others-Agriculture" transition (Pink) is prevalent, likely driven by increasing food demand and reclamation of land.
- **Hydrological Shifts:** The prominent dark brown (Others to Waterbody) and light blue/purple (Waterbody to Others) patches along the western corridor signify active bank erosion and accretion (char formation). This suggests a highly dynamic fluvial environment where land is frequently lost to and gained from the river.

## **4.3 Drivers of Change**

Based on the spatial patterns observed, two primary drivers are hypothesized for the 1995–2005 period:

1. **Anthropogenic Pressure:** The conversion of "Others" to "Agriculture" suggests a transition of natural vegetation or homestead forests into active cropland to support local livelihoods.
2. **Hydro-Geomorphological Processes:** The significant "Waterbody" transitions reflect natural river dynamics, including siltation and course shifting, which reclassify land from terrestrial to aquatic and vice versa (Lambin & Geist, 2006).

## Changes in agricultural land (2015 - 2025)



## Changes in 2015 -2025

| Change                    | Sum of Area_Change | Percentage |
|---------------------------|--------------------|------------|
| Agriculture - Agriculture | 185,417,495.80     | 32.03%     |
| Agriculture - Others      | 90,170,814.95      | 15.58%     |
| Agriculture - Waterbody   | 6,116,801.89       | 1.06%      |
| Others - Agriculture      | 58,650,733.99      | 10.13%     |
| Others - Others           | 184,729,244.90     | 31.91%     |
| Others - Waterbody        | 23,797,084.51      | 4.11%      |
| Waterbody - Agriculture   | 5,912,797.37       | 1.02%      |
| Waterbody - Others        | 24,112,193.40      | 4.17%      |
| Grand Total               | 578,907,166.90     | 100.00%    |

This analysis provides an in-depth breakdown of the land-use transitions between 2015 and 2025, derived from the percentage calculations of the study area. This period reflects a landscape defined by significant stability alongside a vigorous exchange between agricultural and urbanized ("Others") land classes.

### 1. Dominant Stability and Persistence

The data indicates that the vast majority of the landscape remained unchanged throughout the decade. The two largest categories are Agriculture-Agriculture (36.94%) and Others-Others (36.80%). Combined, these figures show that nearly 74% of the total study area maintained its original land-use status. This level of persistence suggests that the core agricultural belts and existing urban/infrastructure centers are relatively well-established, forming the backbone of the region's spatial structure.

### 2. The Agricultural Transition Dynamics

While stability is high, the transitions involving agriculture reveal a more complex story of loss and reclamation:

**Loss to Urbanization/Development:** The most significant shift in the entire dataset is the conversion of Agriculture to Others, which accounts for 17.97% of the land area. This represents a massive outward expansion of development into fertile zones, likely driven by population growth, industrial needs, or infrastructure projects.

**Agricultural Reclamation:** Conversely, 11.69% of the land transitioned from Others to Agriculture. This indicates a notable counter-trend where previously developed or unused land was converted (or returned) to cultivation. However, because the loss to "Others" (17.97%) is significantly higher than the gain (11.69%), the region is experiencing a net "drain" on its agricultural resources.

### 3. Hydrological Transitions

The role of waterbodies in the landscape change is relatively minor but consistent:

**Waterbody Stability:** Only 2.23% of the area remained consistently as water.

**Loss and Gain:** Approximately 4.80% of waterbodies were converted to "Others," while 4.74% of "Others" land was converted back into waterbodies. This near-equal exchange suggests a dynamic water management environment, perhaps involving the filling of old ponds for development while creating new drainage or reservoir systems elsewhere.

**Agriculture-Water Interaction:** The transition between agriculture and water remains the smallest component of change, with only 1.22% of farms becoming water and 1.18% of water being converted to farms.

For the 2015-2025 period, the agricultural sector is the primary stage for land-use conflict. While 36.94% of the area remains safely under cultivation, the high percentage of conversion to "Others" (17.97%) signals a period of rapid spatial transformation. The fact that agriculture is losing more land to development than it is gaining back from other sources suggests a tightening of rural boundaries and a shift toward a more urbanized or industrialized landscape.

The provided land-use map for 2015–2025 offers a spatial visualization of the statistical data previously discussed, illustrating how transitions are geographically distributed across the study area. By cross-referencing the map with the tabular data, we can identify specific regional patterns of agricultural stability and conversion.

### 1. Core Agricultural Zones (Stability)

The map is dominated by a light green shade representing Agriculture-Agriculture, which the tabular data confirms as the largest category at 18,541.75 units (36.94%).

**Spatial Distribution:** This stability is most visible in the northern and central regions of the map, where large, contiguous blocks of land have remained dedicated to farming.

**Context:** These areas represent the resilient agricultural heartland of the study area that has resisted urbanization or conversion over the ten-year period.

### 2. Urban Expansion and Agricultural Encroachment

The vibrant lime-green areas labeled Agriculture-Others represent the most significant land-use shift, totaling 9,017.08 units (17.97%).

**Spatial Distribution:** These transitions are primarily clustered in the southern and south-eastern fringes, as well as pockets in the north-west.

**Interpretation:** This pattern depicts the "fragmentation" of the rural landscape. Development is not occurring in a vacuum but is actively consuming peripheral farmland, often following existing transportation corridors or expanding outward from established "Other" (urban/industrial) hubs.

### 3. Agricultural Reclamation and "Others" Persistence

The map shows a distinct contrast between static urban areas and reclaimed land:

**Stable Non-Agri Land:** The light blue areas (Others-Others) cover 18,472.92 units (36.80%) and are heavily concentrated in the lower-central and southern portions of the map.

**Reclamation (Others-Agriculture):** The pink/mauve clusters represent the 11.69% of land converted back to agriculture. These are scattered throughout the southern half, suggesting that land formerly classified as "Other" is being put back into production, likely through land rehabilitation or rural resettlement programs.

#### 4. Hydrological and Riparian Transitions

The western side of the map features a complex, linear corridor of blues, purples, and pinks, indicating a dynamic river or wetland system.

**Fluvial Dynamics:** The transitions between Waterbody-Others (4.80%) and Others-Waterbody (4.74%) are visually striking along this western artery.

**Agri-Water Interface:** The very small percentages of Agriculture-Waterbody (1.22%) and Waterbody-Agriculture (1.18%) are visible as thin orange and dark green slivers along the edges of these water systems. This shows that while the river system is changing shape, it is not significantly encroaching upon or receding from the primary agricultural belts.

#### 4.4 Change Detection Analysis

A **Post-classification Comparison (PCC)** was conducted to detect changes. By overlaying the maps (e.g., 1985 vs. 2025), a "Transition Matrix" was generated to show exactly how much agricultural land was converted into settlements or lost to the river (Congalton & Green, 2019).

##### **Accuracy Assessment**

Overall Accuracy is the simplest metric. It tells you the proportion of all reference sites that were mapped correctly.

To evaluate the reliability of the land-cover classification, an **accuracy assessment** was conducted using a Confusion Matrix approach. This method allows for a quantitative comparison between the classified land-cover map and reference data, providing measures of both overall accuracy and the agreement between observed and predicted classes.

For historical imagery spanning 1995 to 2005, reference data were obtained from **Google Earth Pro's historical imagery** and verified **topographic maps**. These sources provided spatially accurate information to validate land-cover assignments where field verification was not possible.

$$OA = \frac{\sum_{i=1}^k x_{ii}}{N}$$

| LULC Class        | 2005              |                       | 2015              |                       | 2025              |                       |
|-------------------|-------------------|-----------------------|-------------------|-----------------------|-------------------|-----------------------|
|                   | User Accuracy (%) | Producer Accuracy (%) | User Accuracy (%) | Producer Accuracy (%) | User Accuracy (%) | Producer Accuracy (%) |
| Waterbody         | 77                | 83                    | 82                | 85                    | 93                | 90                    |
| Agriculture       | 84                | 85                    | 85                | 82                    | 85                | 87                    |
| Others            | 79                | 75                    | 87                | 87                    | 92                | 92                    |
| Kappa Coefficient | 78                |                       | 75                |                       | 85                |                       |
| Overall Accuracy  | 80                |                       | 84                |                       | 89                |                       |

Table : Summary of Accuracy Assessment Result

## Kappa Coefficient

The Kappa Coefficient is a more robust statistic because it accounts for the possibility of agreement occurring by chance.

To validate the classification, an accuracy assessment was performed using a **Confusion Matrix**. For historical images (1985–2005), Google Earth Pro historical imagery and topographic maps were used as reference data. For 2025, actual ground-truthing was conducted using a handheld GPS. An overall **Kappa Coefficient** of >0.85 was targeted to ensure reliability.

## Chapter-5: Discussion and Conclusion

### 5.1 Discussion

The landscape evolution over the last three decades tells a story of significant environmental shifts and intensified human activity. When we look at the starting point in 1995, the area was dominated by the "Others" category, which covered more than half of the study area at **58.37%**

(34,479.00 hectares). At that time, agriculture was a secondary feature, occupying about a third of the landscape.

## The 2005 Hydrological Shift

The most striking period of change occurred between 1995 and 2005. The maps in **Image 4** show a dramatic visual expansion of blue tones, particularly along the western river corridors. This visual evidence is backed up by the data in **Image 3**, which shows that **Waterbodies** more than doubled, surging from **7.29%** to a peak of **16.22%**. It is highly likely that this period was marked by significant flooding or a shift in the hydrological regime, as **Agricultural** land simultaneously dropped to its lowest point of **30.93%** during the same decade.

## Agricultural Intensification (2005–2015)

Following the 2005 peak in water coverage, the region saw a massive push toward agricultural reclamation. By 2015, the maps in **Image 5** show a landscape that had turned almost entirely green. The statistics reflect this "green revolution" clearly: **Agricultural** land jumped from 18,268.11 hectares in 2005 to its all-time high of **28,065.33 hectares (47.51%)** in 2015. This period represents the most aggressive transformation in the study area, where both waterbodies and "Others" were converted into productive farmland.

## Stability and Fragmentation (2015–2025)

As we move into the 2025 projection, the landscape appears to be reaching a point of saturation. While **Agriculture** remains a primary land use, it actually saw a slight decrease from its 2015 peak down to **42.53%**. Interestingly, the "Others" category began to rise again, reaching **50.47%** in 2025. The transition matrix in **Image 2** provides a detailed breakdown of this current phase, showing that the "Others-Others" category—land that stayed as 'Others'—is the largest single component of the map, totaling **196,484,715.7** square units. This suggests that while agriculture is widespread, non-agricultural land uses (likely built-up areas or mixed vegetation) are becoming more permanent and consolidated features of the landscape.

## Conclusion

By synthesizing the thirty years of data, we can conclude that the study area has transitioned from a landscape dominated by varied "Other" land types into one heavily defined by agricultural cycles. The most significant long-term trend is the **Agricultural** growth, which expanded at an average annual rate of **0.80%**, representing a net increase of **4,836.87 hectares** since 1995.

The transition matrix for the overall period (1995–2020) in **Image 1** shows that the "Agriculture-Agriculture" class remains the most stable, representing a massive **185,417,495.8** units of land that

did not change. However, the volatility of the region is best exemplified by the waterbodies; while they fluctuate wildly—peaking at **16.22%** in 2005 before settling back to **7.00%** by 2025—they represent a persistent environmental constraint that dictates where other land uses can thrive.

Ultimately, the data suggests that while the region is agriculturally productive, there is a growing trend toward land fragmentation. The recent rise in "Others" (up to **50.47%**) indicates that the landscape is not just becoming a farm, but a complex mosaic of agriculture and other uses. Future land management must account for the hydrological sensitivity of the western corridors and the potential for agricultural land to be slowly encroached upon by the rising "Others" category.

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**Image 1:.** *LULC Transition Area Matrix: 1995–2020.* [Dataset].

**Image 2:.** *LULC Transition Area Matrix: 2015–2025.* [Dataset].

**Image 3:.** *Comparative LULC Classification Statistics: Hectares and Percentages (1995–2025).* [Table].

**Image 4:.** *Spatiotemporal LULC Distribution Maps: 1995 and 2005.* [Figures].

**Image 5:.** *Spatiotemporal LULC Distribution Maps: 2015 and 2025.* [Figures].